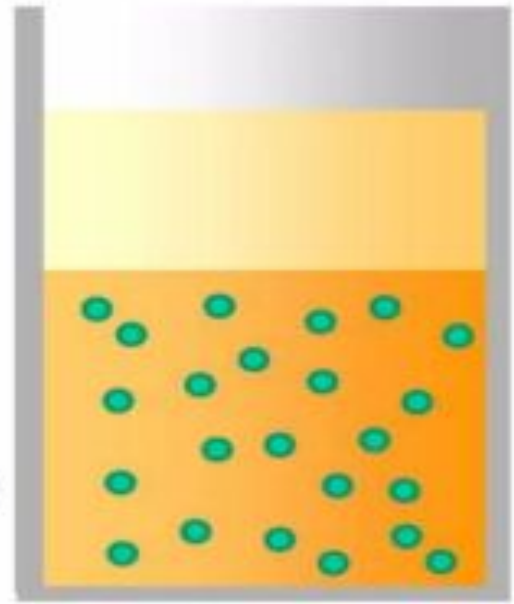
COLLOIDS

- A colloid is a substance microscopically dispersed throughout another substance



- The word colloid comes from a Greek word '**kolla**', which means glue thus colloidal particles are glue like substances.

- These particles pass through a filter pap but not through a semipermeable membrane.

- Colloids can be made settle by the process of centrifugation.

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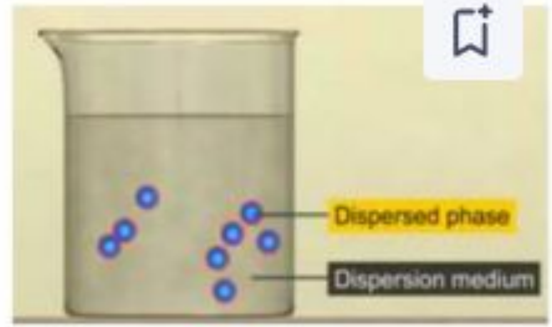
- The colloidal system consist of two phases:

A dispersed phase (A discontinuous phase)

A dispersion medium (A continuous phase)



- The dispersed-phase particles have a diameter of between approximately 1nm – 100nm .
- Such particles are normally invisible in an optical, though their presence can be confirmed with the use of an ultramicroscope or an electron microscope.

Solutions

- Made up of particles or solutes and a solvent
- The solvent part of the solution is usually a liquid, But can be a gas.
- The particles are atoms, ions or molecules that Are very small in diameter.

Colloidal Mixture

- Has particles that are not as small as a solution And not as large as a suspension.
- The particles are intermediate in size.

Suspensions

- Made up of particles and a solvent. Its particles are larger than those found in a solution.
- The particles in a suspension can be distributed throughout the suspension evenly by Shaking the mixture.

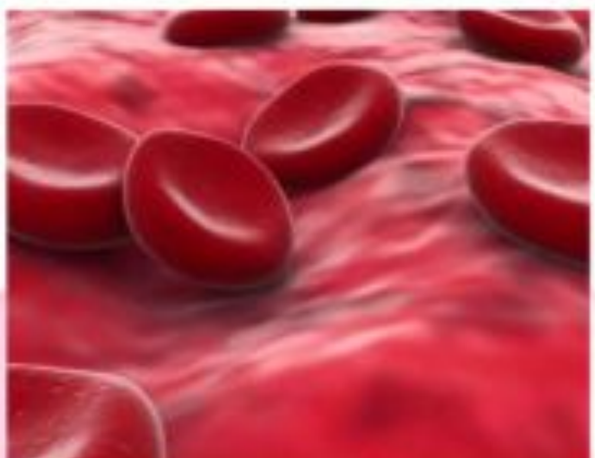

Solution:
Table salt dissolves in water to form Salt (saline) water. A solute (salt; NaCl)Is dissolved in another substance (water) known as a solvent, and this creates a solution.
Suspension:
Flour suspended in water (appears light blue because blue light is scattered off the flour particles to a greater extent than red light)
Colloid:
Milk is an emulsified colloid of liquid butterfat globules dispersed within a water based liquid. Colloids are Stabilized In suspension by Electrostatics - mutual Repulsion of like electrical Charges.
www.ToleranceLost.com

Comparison of the Properties of Solutions, Colloids, And Suspensions

Property	True Solution	Colloid	Suspension
Particle Size	Less than 1 nm	1 to 100 nm	More than 100 nm
Appearance	Clear	Cloudy	Cloudy
Homogeneity	Homogeneous	Homogeneous or Heterogeneous	Heterogeneous
Transparency	Transparent but often coloured	Often translucent and opaque but can be transparent	Often opaque but can be translucent
Separation	Does not separate	Can be seperated	Separates or settles
Filterability	Passes through filter paper	Passes through filter paper	Particles do not pass through filter paper

- Examples of colloids are milk, synthetic polymers, fog, blood, jam, shoe polish, smoke, etc.




CLASSIFICATION OF COLLOIDS

- Based of physical state of dispersed phase an dispersion medium.
- Based of nature of interaction between dispersed phase and dispersion medium.
- Based on molecular size in the dispersed phase.
- Based on appearance of colloids.
- Based on electric charge on dispersion phase.

Types of Colloidal Systems

Type Name	Dispersed Phase	Dispersion medium	Examples
Foam	gas	liquid	whipped cream, shaving cream, soda-water
Solid foam	gas	solid	froth cork, pumice stone, foam rubber
Aerosol	liquid	gas	for, mist, clouds
Emulsion	liquid	liquid	milk, hair cream
Solid emulsion (gel)	liquid	solid	butter, cheese
Smoke	solid	gas	dust, soot in air
Sol	solid	liquid	paint, ink, colloidal gold
Solid sol	solid	solid	ruby glass (gold dispersed in glass), alloys.

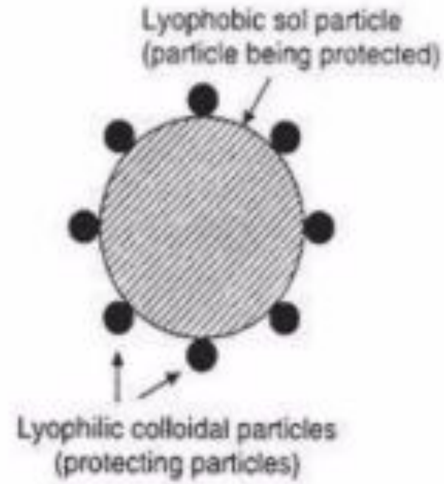
BASED ON NATURE OF INTERACTION BETWEEN DISPERSED PHASE AND DISPERSION MEDIUM

LYOPHILIC COLLOIDS

- Colloidal solution in which the dispersed phase has a great affinity for the dispersion medium.
- They are also termed as **intrinsic colloids**.
- Such substances have tendency to pass into colloidal solution when brought in contact with dispersion medium.
- If the dispersion medium is water, they are called hydrophilic or emulsoids.
- The lyophilic colloids are generally self- stabilized.
- Reversible in nature and are heavily hydrated.
- Example of lyophilic colloids are starch, gelatin, rubber, protein etc.

LYOPHOBIC COLLOIDS

- Colloidal solutions in which the dispersed phase has no affinity to the dispersion medium.
- These are also referred as **extrinsic colloids**.
- Such substances have no tendency to pass into colloidal solution when brought in contact with dispersion medium.
- The lyophobic colloids are relatively unstable.
- They are irreversible by nature and are stabilized by adding small amount of electrolyte.
- They are poorly hydrated.
- If the dispersion medium is water, the lyophobic colloids are called hyrophobic or suspenoids.
- Examples: sols of metals like Au, Ag, sols of metal hyroxides and sols of metal sulphides.



PREPARATION OF SOLS

Lyophilic sols may be prepared by simply warming the solid with the liquid dispersion medium *e.g.*, starch with water. On the other hand, lyophobic sols have to be prepared by special methods. These methods fall into two categories :

(*a*) Dispersion Methods in which larger macro-sized particles are broken down to colloidal size.

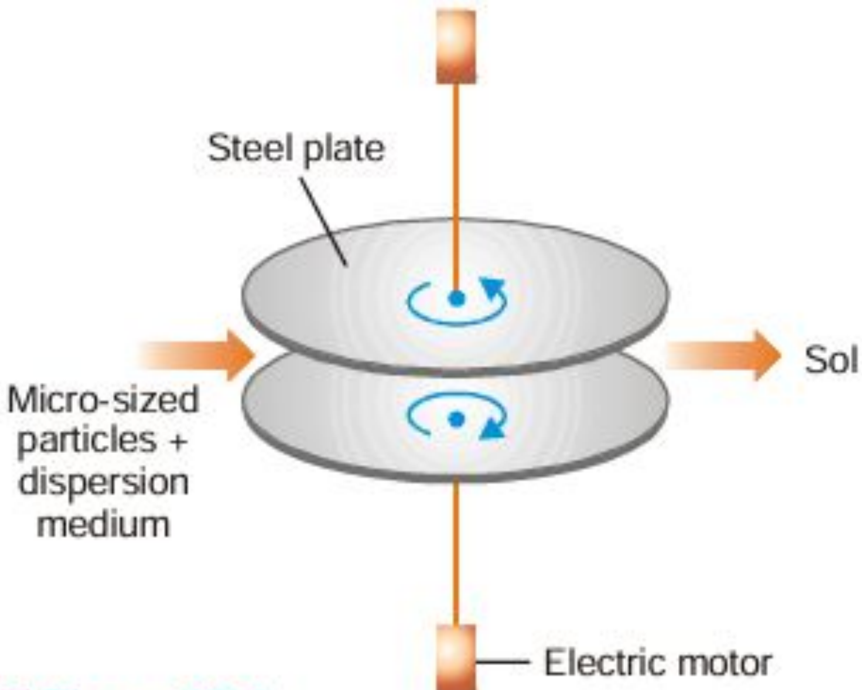
(*b*) Aggregation Methods in which colloidal size particles are built up by aggregating single ions or molecules.

DISPERSION METHODS

In these methods, material in bulk is dispersed in another medium.

(1) Mechanical dispersion using Colloid mill

The solid along with the liquid dispersion medium is fed into a Colloid mill. The mill consists of two steel plates nearly touching each other and rotating in opposite directions with high speed. The solid particles are ground down to colloidal size and are then dispersed in the liquid to give the sol. 'Colloidal graphite' (a lubricant) and printing inks are made by this method.



■ **Figure 22.4**
A Colloid mill.

(2) Bredig's Arc Method

It is used for preparing hydrosols of metals *e.g.*, silver, gold and platinum. An arc is struck between the two metal electrodes held close together beneath *de-ionized* water. The water is kept cold by immersing the container in ice/water bath and a trace of alkali (KOH) is added. The intense heat of the spark across the electrodes vaporises some of the metal and the vapour condenses under water. Thus the atoms of the metal present in the vapour aggregate to form colloidal particles in water. Since the metal has been ultimately converted into sol particles (*via* metal vapour), this method has been treated as of dispersion.

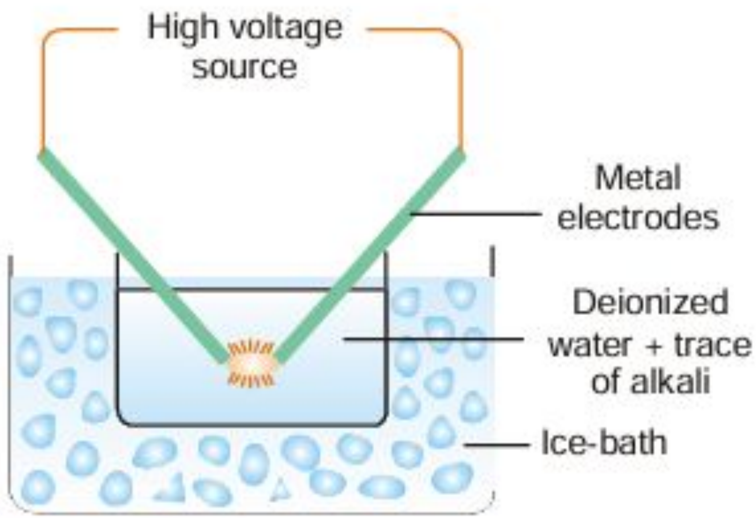


Figure 22.5
Bredig's Arc method.

Non-metal sols can be made by suspending coarse particles of the substance in the dispersion medium and striking an arc between iron electrodes.

(3) By Peptization

Some freshly precipitated ionic solids are dispersed into colloidal solution in water by the addition of small quantities of electrolytes, particularly those containing a common ion. The precipitate adsorbs the common ions and electrically charged particles then split from the precipitate as colloidal particles.

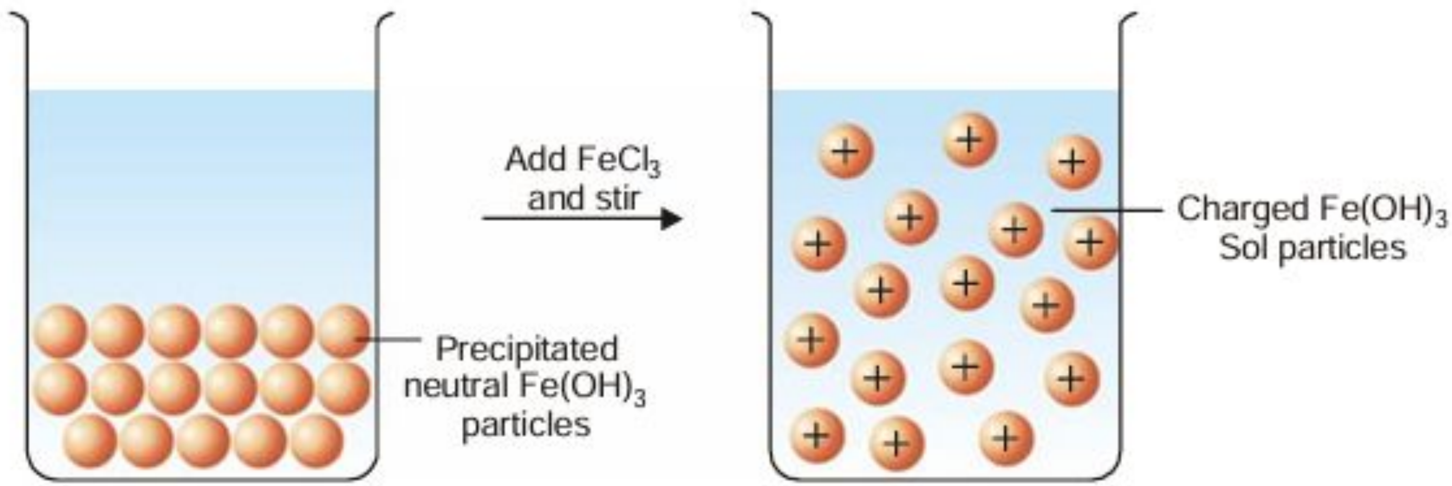


Figure 22.6

Sol of ferric hydroxide is obtained by stirring fresh precipitate of ferric hydroxide with a small amount of $FeCl_3$.

The dispersal of a precipitated material into colloidal solution by the action of an electrolyte in solution, is termed peptization. The electrolyte used is called a peptizing agent.

Purification of colloids

There are three common methods used for purification of colloids.

- Dialysis
- Electrodialysis
- Ultra filteration

Dialysis

Animal membranes (bladder) or those made of parchment paper and cellophane sheet, have very fine pores. These pores permit ions (or small molecules) to pass through but not the large colloidal particles. When a sol containing dissolved ions (electrolyte) or molecules is placed in a bag of permeable membrane dipping in pure water, the ions diffuse through the membrane. By using a continuous flow of fresh water, the concentration of the electrolyte outside the membrane tends to be zero. Thus diffusion of the ions into pure water remains brisk all the time. In this way, practically all the electrolyte present in the sol can be removed easily.

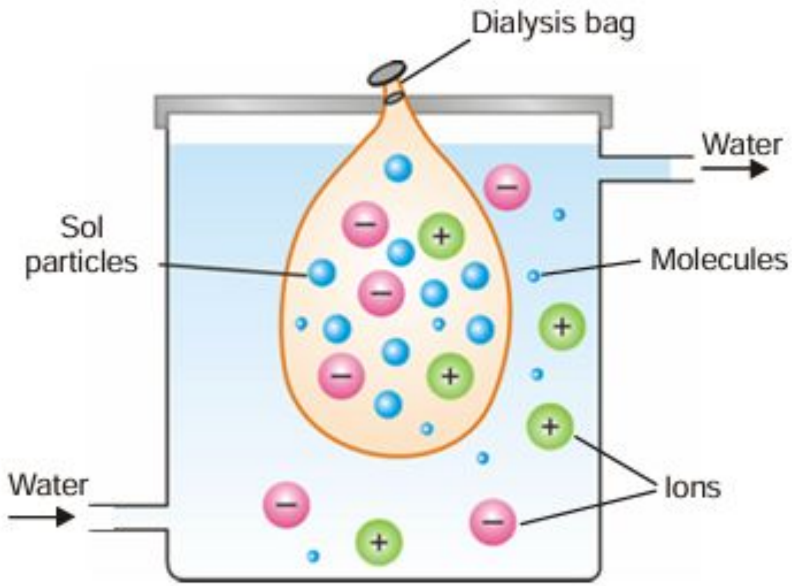


Figure 22.7
Dialysis of a sol containing ions and molecules.

The process of removing ions (or molecules) from a sol by diffusion through a permeable membrane is called Dialysis. The apparatus used for dialysis is called a **Dialyser.**

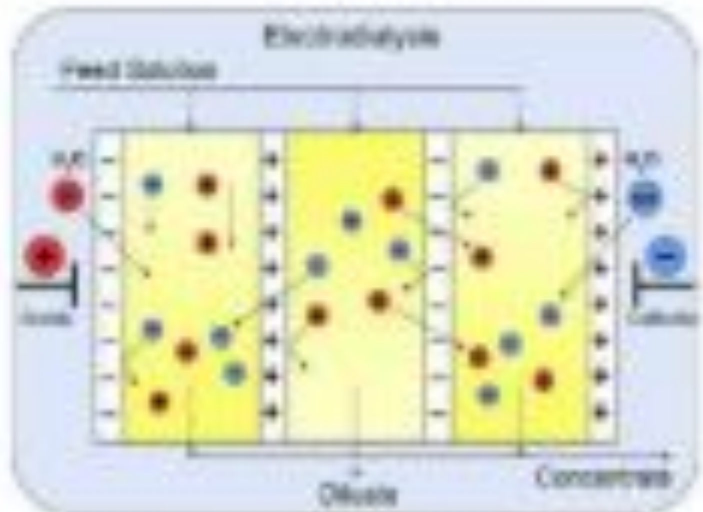
Example. A ferric hydroxide sol (red) made by the hydrolysis of ferric chloride will be mixed with some hydrochloric acid. If the impure sol is placed in the dialysis bag for some time, the outside water will give a white precipitate with silver nitrate. After a pretty long time, it will be found that almost the whole of hydrochloric acid has been removed and the pure red sol is left in the dialyser bag.

Electro dialysis

➢The process of dialysis is very slow.

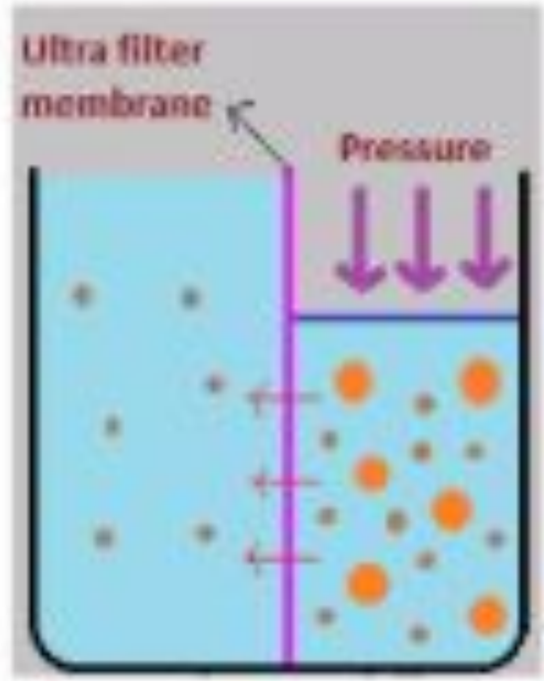
➢The process is speeded up by application of electrical potential.

➢This is called electro dialysis.



Ultra filtration

- Ultra filtration is a process of high pressure
  filtration through a semi permeable membrane
  in which colloidal particles are retained while
  the small sized solutes and the solvent are
  forced to move across the membrane by
  hydrostatic pressure forces.



PHYSICAL PROPERTIES OF COLLOIDS

- **Heterogeneity:** Colloidal solutions consist of two phases-dispersed phase and dispersion medium.
- **Visibility of dispersed particles:** The dispersed particles present in them are not visible to the naked eye and they appear homogenous.
- **Filterability:** The colloidal particles pass through an ordinary filter paper. However, they can be retained by animal membranes, cellophane membrane and ultrafilters.
- **Stability:** Lyophilic sols in general and lyophobic sols in the absence of substantial concentrations of electrolytes are quite stable.
- **Colour:** The colour of a colloidal solution depends upon the size of colloidal particles present in it. Larger particles absorb the light of longer wavelength and therefore transmit light of shorter wavelength.

OPTICAL PROPERTIES OF SOLS

(1) Sols exhibit Tyndall effect

When a strong beam of light is passed through a sol and viewed at right angles, the path of light shows up as a hazy beam or cone. This is due to the fact that sol particles absorb light energy and then emit it in all directions in space. This 'scattering of light', as it is called, illuminates the path of the beam in the colloidal dispersion.

The phenomenon of the scattering of light by the sol particles is called Tyndall effect.

The illuminated beam or cone formed by the scattering of light by the sol particles is often referred as **Tyndall beam** or **Tyndall cone.**

The hazy illumination of the light beam from the film projector in a smoke-filled theatre or the light beams from the headlights of car on a dusty road, are familiar examples of the Tyndall effect. If the sol particles are large enough, the sol may even appear turbid in ordinary light as a result of Tyndall scattering.

True solutions do not show Tyndall effect. Since ions or solute molecules are too small to scatter light, the beam of light passing through a true solution is not visible when viewed from the side. Thus Tyndall effect can be used to distinguish a colloidal solution from a true solution.

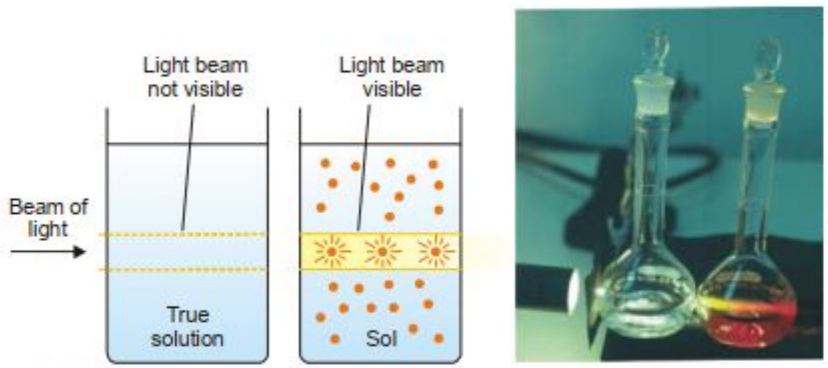


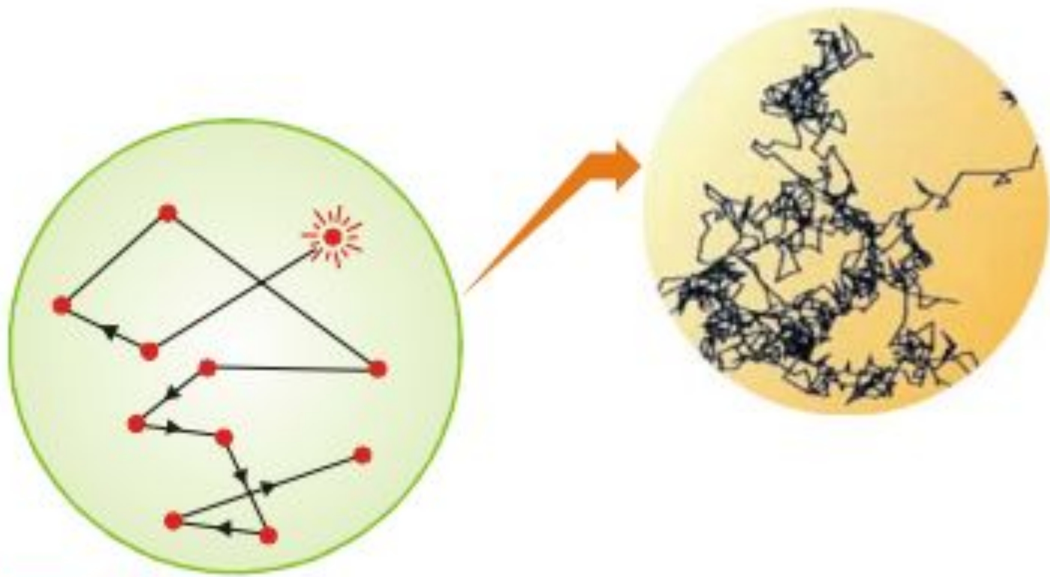
Figure 22.10
Tyndall effect (Illustration).

KINETIC PROPERTIES OF SOLS

Brownian Movement

When a sol is examined with an ultramicroscope, the suspended particles are seen as shining specks of light. By following an individual particle it is observed that the particle is undergoing a constant rapid motion. It moves in a series of short straight-line paths in the medium, changing directions abruptly.

The continuous rapid zig-zag movement executed by a colloidal particle in the dispersion medium is called Brownian movement or motion.



■ Figure 22.13
An illustration of Brownian movement.

This phenomenon is so named after Sir Robert Brown who discovered it in 1827.

Suspension and true solutions do not exhibit Brownian movement.

Electrical Double layer

The surface of colloidal particle acquires a positive charge by selective adsorption of a layer of positive ions around it. This layer attracts counterions from the medium which form a second layer of negative charges. The combination of the two layer of +ve and –ve charges around the sol particle was called **Helmholtz Double layer.** Helmholtz thought that positive charges next to the particle surface were fixed, while the layer of negative charges along with the medium were mobile.

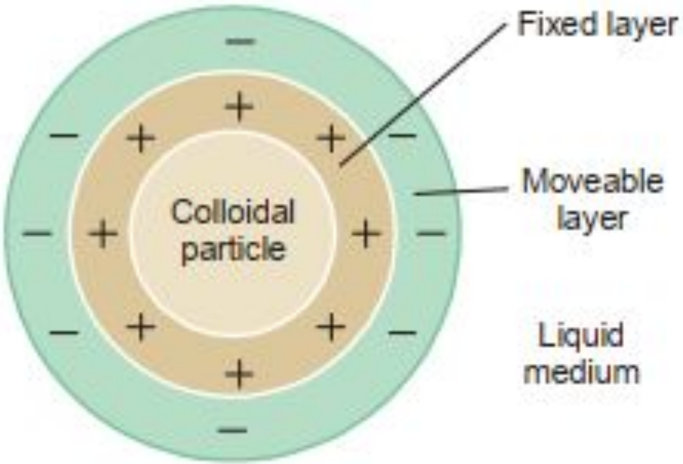


Figure 22.17
Helmholtz double layer.

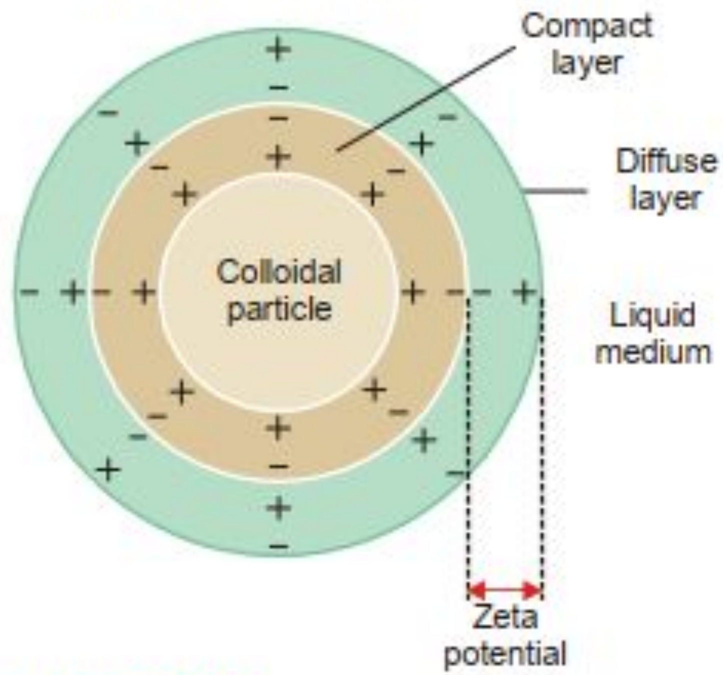


Figure 22.18
The electrical double layer (Stern).

More recent considerations have shown that the double layer is made of :

(*a*) a **Compact layer** of positive and negative charges which are fixed firmly on the particle surface.

(*b*) a **Diffuse layer** of counterions (negative ions) diffused into the medium containing positive ions.

(4) Coagulation or Precipitation

We know that the stability of a lyophobic sol is due to the adsorption of positive or negative ions by the dispersed particles. The repulsive forces between the charged particles do not allow them to settle. If, some how, the charge is removed, there is nothing to keep the particles apart from each other. They aggregate (or flocculate) and settle down under the action of gravity.

The flocculation and settling down of the discharged sol particles is called coagulation or precipitation of the sol.

How coagulation can be brought about?

The coagulation or precipitation of a given sol can be brought about in four ways :

(*a*) By addition of electrolytes

(*b*) By electrophoresis

(*c*) By mixing two oppositely charged sols

(*d*) By boiling

EMULSIONS

These are liquid-liquid colloidal systems. In other words, **an emulsion may be defined as a dispersion of finely divided liquid droplets in another liquid.**

Generally one of the two liquids is *water* and the other, which is immiscible with water, is designated as *oil*. Either liquid can constitute the dispersed phase.

Types of Emulsions

There are two types of emulsions.

(*a*) **Oil-in-Water type (O/W type)** ; (*b*) **Water-in-Oil type (W/O type)**

ADVANTAGES OF COLLOIDS

- Colloids allow the dispersion of normally insoluble materials, such as metallic gold or fats. These can then be used more easily, or absorbed more easily.

- Colloidal gold, for example, can be used in medicine to carry drugs and antibiotics, because it is highly non-reactive and non-toxic.

- Pharmaceutical industry makes use of colloidal solution preparation in many medicines. A wide variety of medicines are emulsions. An example is Cod Liver Oil.

- Paint industry also uses colloids in the preparation of paints.

- In milk, the colloidal suspension of the fats prevents the milk from being thick, and allows for easy absorption of the nutrients.

- Sewage water contains particles of dirt, mud etc. which are colloidal in nature and carry some electrical charge. These particles may be removed by using the phenomenon of electrophoresis.

- The sky is the empty space around earth and as such has no colour. It appears blue due to the scattering of light by the colloidal dust particles present in air (Tyndall effect).

- Asphalt emulsified in water and is used for building roads.

➢The sugar present in milk produces lactic acid on fermentation. Ions produced by acid, destroy the charge on the colloidal particles present in milk, which then coagulate and separate as curd.

➢Soap solution is colloidal in nature. It removes the dirt particles either by adsorption or by emulsifying the greasy matter sticking to the cloth.

➢Large numbers of food particles which we use in our daily life are colloidal in nature.

Example: Milk, butter, & ice cream etc


COLLOIDAL GOLD
CHERRY
BLOSSOM
SHOE POLISH
BLACK
glue
Colloidal Silver
10 ppm